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TECHNOLOGICAL UNIVERSITY DUBLIN

CITY CAMPUS - GRANGEGORMAN

TU856 BSc. (Honours) Degree in Computer Science
TU857 BSc. (Honours) Degree in Computer Science (Infrastructure)
TU858 BSc. (Honours) Degree in Computer Science (International)

Year 4

SEMESTER 1 EXAMINATIONS 2024/25

CMPU4011 Machine learning for Data Analytics

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Instructions To Candidates:

ANSWER ALL QUESTIONS.

Exam Duration: 2 hours

1. (a) Explain what is meant by a machine learning model being **consistent** with the data. Discuss if the model being consistent is a good or bad thing. (10 marks)
- (b) Discuss what is meant by the **curse of dimensionality** in the context of data analytics, and suggest how we can deal with it. (10 marks)
- (c) Discuss the use of a confusion matrix when evaluating machine learning algorithms, and what advantages a confusion matrix might have over using a misclassification rate. (10 marks)
- (d) In the context of machine learning, explain the concept of **cross-validation**, and distinguish between **k-fold** cross validation and **leave-one-out** cross validation, listing a possible use case for each. (10 marks)
2. (a) In the context of machine learning, explain what is the **inductive bias** for
- the k-NN algorithm and for
 - ID3 algorithm. (10 marks)
- (b) Discuss the difference between **supervised** and **unsupervised** machine learning. (5 marks)
- (c) Table 1 below lists a small dataset of five customers and whether they have taken a premium subscription.

Age (years)	Monthly Spend (€)	Website Visits (per month)	Premium subscription (Yes/No)
22	30	5	No
34	100	12	Yes
28	60	8	No
45	150	25	Yes
60	200	10	Yes

Table 1: Dataset of customer details showing whether they have a premium subscription or not.

Explain what prediction the k-Nearest Neighbor algorithm (where k=3) will make for a new customer of age 30, monthly spend 50 and 10 website visits per month. Include the intermediate steps in your answer, not just the final prediction.

(10 marks)

(Question 2 continues on the next page.)

(d) Discuss the role of **basis functions** in the context of linear regression machine learning models.

(5 marks)

3. (a) Explain how a decision tree algorithm determines the best feature to split on at each node. In your answer, describe the concept of **entropy** and how it is used to guide the algorithm.

(10 marks)

(b) **Overfitting** can be a problem when building predictive models. Briefly explain how this can be address in decision tree models.

(5 marks)

(c) Briefly outline the difference between **linear regression** and **logistic regression** models.

(5 marks)

(d) You have been asked to build a classifier that will predict whether a customer should be approved for a car loan. You have been provided with lists of individuals who have already applied for a car loan along with the outcome, whether they were given a loan or not.

You have been told that the key characteristics of the individual that are typically used to determine the outcome include age, their employment status (i.e. employed or not), and whether they are a home owner or not.

Describe how you will build a Naive Bayes classifier which can be used to predict the outcome of a car loan application.

(10 marks)